Daily climate data for Europe

Short documentation (README)

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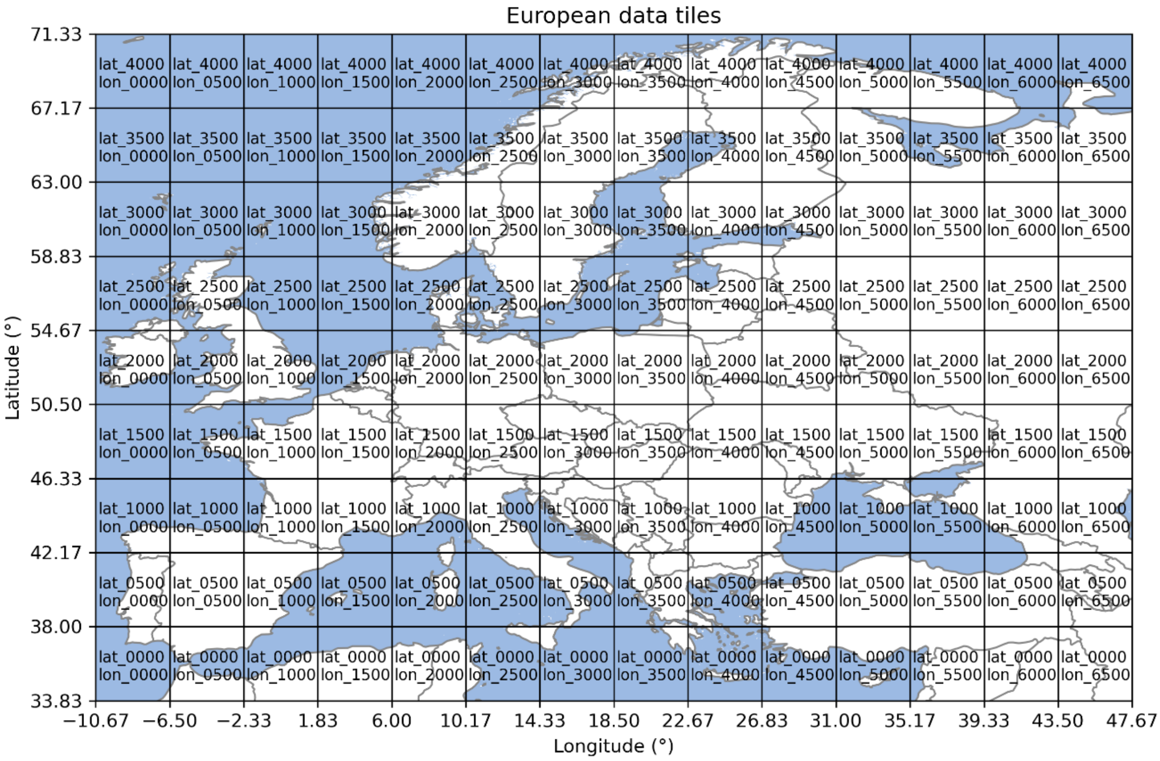

1. Historical

- **Resolution**: 30x30 arcsec
- **Temporal coverage**: 01.01.1950-30.11.2022
- **Extent**: 10.67°W – 47.67°E, 33.68°N – 71.33°N
- Daily data divided into **tiles** of 500x500 grid cells
- **Variables**: Temperature (tas, tasmin, tasmax), Precipitation (pr), Vapor pressure deficit (vpd), mean wind speed (wspd), daily maximum wind gust (wsgsmax), potential evapotranspiration (PET), global radiation (rsds)
- **Data sources:**
    - CHELSA-W5E5 v1.1 (Karger D. N., 2023) for variables precipitation (pr), global radiation (rsds), mean temperature (tas), maximum temperature (tasmax) and minimum temperature (tasmin)
    - CHELSA V2.1 (Karger D. N.-A., 2017): for climatological average monthly wind speed (sfcWind_01, …, sfcWind_12) and for climatological mean vapor pressure deficit (vpd_01, …, vpd_12)
    - ERA5-Land (Munoz-Sabater, 2021): for variables preciptation (pr), global radiation (rsds), mean temperature (tas), dew point (tds), and wind speed (sfcWind)
    - ERA5 (Hersbach, 2020): Global radiation (ssrd), maximal vertical wind speed, CAPE, horizontal wind speed, and wind (550/750 hPa)

Methods

i. tas, tasmin, tasmax, pr, rsds

For those variables for which daily data from CHELSA exist, a statistical bias adjustment of ERA5-Land was carried out in order to obtain a homogeneous data set for 1951 to 2022. The correction was made using 1981-2010 as the observation period with the CHELSA- W5E5 v1.1 data using the "Quantile Delta Mapping" (QDM) methodology according to (Cannon, 2015) (Lehner, 2023).

In CHELSA-W5E5 v1.1, the variable rsds ('Surface Downwelling Shortwave Radiation') represents the radiation on the real surface, estimating from the range of the values. The radiation on the horizontal surface was created with a statistical approach using data from Austria (see internal report in FORSITE II).

ii. wspd

Only climatological monthly averages of wind speed are available for the higher-resolution CHELSA data set. An approach was chosen in which the temporal variability and trends of ERA5-Land are preserved and the climatological means of CHELSA V2.1 are reproduced at the same time:

- Calculation of climatological monthly means for the period 1981-2010 for ERA5-Land
- Calculation of the ratio of CHELSA climatology to ERA5-Land climatology for each of the 12 months.
- Multiplication of the daily values with the climatological ratio of the CHELSA climatology to the ERA5-Land climatology.

iii. vpd

Only climatological monthly averages exist for the higher-resolution CHELSA data set. However, a random validation of the vapor pressure deficit of CHELSA V2.1 showed that the values deviate significantly from the measured values in Austria, in some cases by 50% in the monthly mean. Therefore, an approach had to be chosen in which not only the temporal variability and trends of ERA5-Land are retained, but also spatial mean values, as these are closer to the measured values in Austria. This means that only the small-scale spatial variability of CHELSA is largely retained. This required the following steps:

- Conversion of the dew point of ERA5-Land into the vapor pressure deficit using temperature
- Calculation of climatological monthly means of the vapor pressure deficit for the period 1981-2010 for ERA5-Land
- Calculation of the ratio of CHELSA climatology to ERA5-Land climatology for each of the 12 months.
- Smoothing of the ratio with a running mean in both spatial dimensions with a centered window of size 0.1°. This brings the ratio of the two climatologies approximately to the real resolution of ERA5-Land.
- Multiplication of the daily values by the climatological ratio of the CHELSA climatology to the ERA5 land climatology, divided by the smoothed ratio.

iv. PET

Penman-Monteith evapotranspiration (FAO-56) (Zotarelli, 2010).

v. wsgsmax

Multi-stage approach was conducted to generate gridded wind gust data for the European domain that covers both historical and future periods.

Initially, a 1x1 km resolution gridded wind gust data set for Austria for the historical time period was produced. Data from the years 1981 to 2012 were derived using available gridded INCA dataset provided by GeoSphere Austria (2012-2021). A Random Forest Regressor model was applied over each grid cell using the following set of predictors: mean wind speed, tasmax, tasmin, tds, sin & cos (represent annual cycle), pr, maximal vertical wind speed, CAPE, horizontal wind speed (represents turbulence), and wind shear below 6km (550/750 hPa). The last four predictors were derived from ERA5 reanalysis. Subsequently, to address inhomogeneity between datasets, Quantile Mapping was employed for bias adjustment.

In the second stage, a gridded wind gust dataset for Europe was produced. A model trained on the Austrian domain (which corresponds to two tiles) was applied to Europe. Two models were implemented to correctly capture and predict extreme wind gust values. The feature matrix used for the models includes the following features: sin & cos, DEM, exp_5km, exp_10km, mean wind speed, gust factor (ERA5 wind gust/mean wind speed), pr, and diurnal range (tasmax – tasmin). The first model, a Random Forest Classifier, calculates probabilities of events, assigning P=1 to (extreme) events above the 85th percentile or P=0 to the rest. The second model, a Random Forest Quantile Regressor (RFQR), predicts 50 to 90th percentiles, thereby focusing on the distribution's tail. The final wind gust was determined by the formula:

RFQR_50 x (1-P) + RFQR_90 x (P)

2. Scenario data

- **Resolution**: 30x30 arcsec
- **Temporal coverage**: 01.01.1991-31.12.2100
- **Extent**: 10.67°W – **39.33°**E, 33.68°N – 71.33°N (smaller than historical)
- Daily data divided into **tiles** of 500x500 grid cells
- **Variables**: Temperature (tas, tasmin, tasmax), Precipitation (pr), Vapor pressure deficit (vpd), mean wind speed (wspd), daily maximum wind gust (wsgsmax), potential evapotranspiration (PET), global radiation (rsds)
- **Data sources**:
  - Historical data (for bias adjustment)
  - Three EUROCORDEX climate models (Jacob, 2014), representative for three Global Warming Levels (GWL, °C):
    - GWL<2.0: MPI-M-MPI-ESM-LR_rcp45_r1i1p1_CLMcom-CCLM4-8-17
    - GWL3.0: MPI-M-MPI-ESM-LR_rcp85_r1i1p1_CLMcom-CCLM4-8-17
    - GWL4.0: ICHEC-EC-EARTH_rcp85_r12i1p1_SMHI-RCA4

Methods

The scenario data was bias adjusted and downscaled using the approaches described in (Lehner, 2023). The historical 30-year climate period 1991-2020 was selected for bias adjustment. The climate model data were interpolated to the same spatial resolution ("patch" method). The climate models are corrected in 30-year periods whereby the period is always shifted by 10 years (i.e. 2011-2040, 2021-2050, 2031-2060, etc.). Only the middle 10 years are stored (2021-2030, 2031-2040, 2041-2050, etc.). Depending on the variable, the bias adjustment was either additive or multiplicative, i.e. model errors were corrected either relatively or absolutely. The temperature (minimum, mean, maximum) and humidity (dew point) are adjusted additively. For all other variables (wind speed, global radiation, precipitation) the correction was carried out multiplicatively.

For the future scenario runs of wind gusts the two aforementioned models were used, with the same initial settings.

3. Literature

Cannon, A. J. (2015). Bias correction of GCM precipitation by quantile mapping: how well do methods preserve changes in quantiles and extremes?. *Journal of Climate*, pp. 6938-6959.

Hersbach, H. a.-S. (2020). The ERA5 global reanalysis. *Wiley Online Library*, pp. 1999-2049.

Jacob, D. P. (2014). EURO-CORDEX: new high-resolution climate change projections for European impact research. *Reg Environ Change* , pp. 563–578.

Karger, D. N. (2023). CHELSA-W5E5: Daily 1 km meteorological forcing data for climate impact studies. *Earth System Science Data Discussions*, pp. 1-28.

Karger, D. N.-A. (2017). Climatologies at high resolution for the earth's land surface areas. *Nature Publishing Group*, pp. 1-20.

Lehner, F. a. (2023). Evaluating skills and issues of quantile-based bias adjustment for climate change scenarios. *Advances in Statistical Climatology*, *Meteorology and Oceanography*, pp. 29-44.

Munoz-Sabater, J. a.-P. (2021). ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth system science data*, pp. 4349-4383.

Zotarelli, L. a. (2010). Step by step calculation of the Penman-Monteith Evapotranspiration (FAO-56 Method). *Institute of Food and Agricultural Sciences. University of Florida*.